A3 Supervised Learning

Artificial Intelligence
AIN2IL (MTD.ba)
SS2026

Submission via Moodle **40 Points**

Exercise 1 (20 points) Analyze data set 1 (available for download on Moodle in the assignment). The last column is the label column, the other columns are the input features. Put aside a test set with 30% test samples and proceed with the remaining 70%. Use the following machine learning methods along with hyperparameter optimization on this data set:

- k-nearest neighbor with several choices of k
- SVM with RBF kernel and several choices of C and γ
- Random forest (with different depths of trees; use a fixed number of 1000 trees)

In order to estimate the generalization error, use grid search with 10-fold cross validation to determine the average classification error for each method. Also compute the final results on the test set.

Exercise 2 (20 points) Repeat your analysis from Exercise 1 in two ways: (1) after flipping labels randomly with a probability of 0.2, (2) after adding fifteen noise features with random numbers that are uniformly distributed on the unit interval $[0, 1]$. What do you observe? Try to explain possible changes of the generalization error and the best hyperparameters.

Submit your program code, along with short reports that summarize your results, visualizations, and interpretations. This should all be done inside notebooks. Hand in the notebook file(s) and HTML dump(s).